General Relativity: Proper Distance, Kruskal Geometry, and Horizon Observers

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Chapter 1

Proper Distance, Kruskal Geometry, and Horizon Observers

The Schwarzschild metric appears to become singular at the horizon. The curvature does not. To separate those two statements, we first remove the overall size of the black hole, then replace the radial coordinate by proper distance from the horizon. Near the horizon the resulting geometry is ordinary flat spacetime written in accelerated coordinates. Once that geometry is drawn, questions about Alice, Bob, and Charlie become questions about worldlines and light rays rather than verbal paradoxes.

1.1 Scaling Out the Schwarzschild Radius

Begin with primed coordinates so that the rescaled variables can later carry the familiar names:

$$d\tau^2 = \left(1 - \frac{2MG}{r'}\right) dt'^2 - \frac{dr'^2}{1 - \frac{2MG}{r'}} - r'^2 d\Omega_2^2. \tag{1.1}$$

Introduce the Schwarzschild radius and dimensionless coordinates

$$r_{\mathrm{S}} = 2MG, \qquad r = \frac{r'}{r_{\mathrm{S}}}, \qquad t = \frac{t'}{r_{\mathrm{S}}}. \tag{1.2}$$

We use $c = 1$, so time and length have the same units. The radius r_{S} is therefore both a length and the light-crossing time associated with that length. Substitution gives

$$d\tau^2 = r_{\mathrm{S}}^2 \left[\left(1 - \frac{1}{r}\right) dt^2 - \frac{dr^2}{1 - \frac{1}{r}} - r^2 d\Omega_2^2 \right]. \tag{1.3}$$

All nonrotating, uncharged Schwarzschild black holes consequently have the same dimensionless geometry. Changing the mass changes the ruler multiplying the metric, not the shape of the expression in brackets. The situation is like drawing spheres after dividing every distance by the sphere's radius: the same drawing describes the Moon and the Earth, although a centimeter on one drawing represents a different physical distance.

Question from the audience. Which signs belong to the radial and angular pieces of the metric?

Response. With the $+---$ signature used here, both spatial pieces carry minus signs. The room caught the sign while the expression was being written. The angular factor may use a sine or cosine according to whether the polar coordinate is measured from the axis or from the equator; the compact notation $d\Omega_2^2$ avoids that convention issue. [a]

[a] Lecture timestamp: 00:07:48–00:08:12.

1.2 The Curvature Scale at the Horizon

The rescaling answers an important physical question without a long tensor calculation. A metric component is dimensionless. A Christoffel symbol contains one coordinate derivative and therefore has the dimensions of inverse length, while a component of the Riemann tensor in a physical orthonormal frame has the dimensions

$$[g_{\mu\nu}] = 1, \qquad [\Gamma^{\mu}{}_{\nu\rho}] = L^{-1}, \qquad [R_{\hat{\alpha}\hat{\beta}\hat{\gamma}\hat{\delta}}] = L^{-2}. \tag{1.4}$$

At the horizon $r = 1$, the only available physical length is r_{S}. The tidal-curvature scale must therefore be

$$\left| R_{\hat{\alpha}\hat{\beta}\hat{\gamma}\hat{\delta}} \right|_{\mathrm{hor}} \sim \frac{1}{r_{\mathrm{S}}^2}. \tag{1.5}$$

More generally it is r_{S}^{-2} times a dimensionless function of r. This statement concerns the Riemann tidal field, not the Ricci scalar, which vanishes in the Schwarzschild vacuum.

The physical conclusion is immediate. Increasing the mass increases r_{S} and weakens the tidal field at the horizon. A small black hole can pull an extended body apart while it crosses; near the horizon of a supermassive black hole, tidal forces can be mild. Curvature nevertheless grows inward and diverges at $r = 0$. The horizon and the singularity are therefore radically different objects.

1.3 Proper Distance from the Horizon

For the rest of the local calculation, set $r_{\mathrm{S}} = 1$. Coordinate changes are not cosmetic in relativity: a poor coordinate can hide a regular geometry behind singular coefficients. The Schwarzschild coordinate difference $r - 1$ is not the measured radial distance from the horizon.

Hold t and the angles fixed. The positive spatial line element is then

$$ds^2 = \frac{dr^2}{1 - \frac{1}{r}}, \qquad ds = \sqrt{\frac{r}{r-1}}\, dr. \tag{1.6}$$

Define the dimensionless proper distance from the horizon by

$$\rho(r) = \int_1^r \sqrt{\frac{r'}{r'-1}}\, dr'. \tag{1.7}$$

The physical distance is $r_{\mathrm{S}}\rho$ when the overall scale is restored. We shall not need the closed form of the integral. We need only its derivative:

$$\frac{d\rho}{dr} = \sqrt{\frac{r}{r-1}}, \qquad d\rho^2 = \frac{dr^2}{1 - \frac{1}{r}}. \tag{1.8}$$

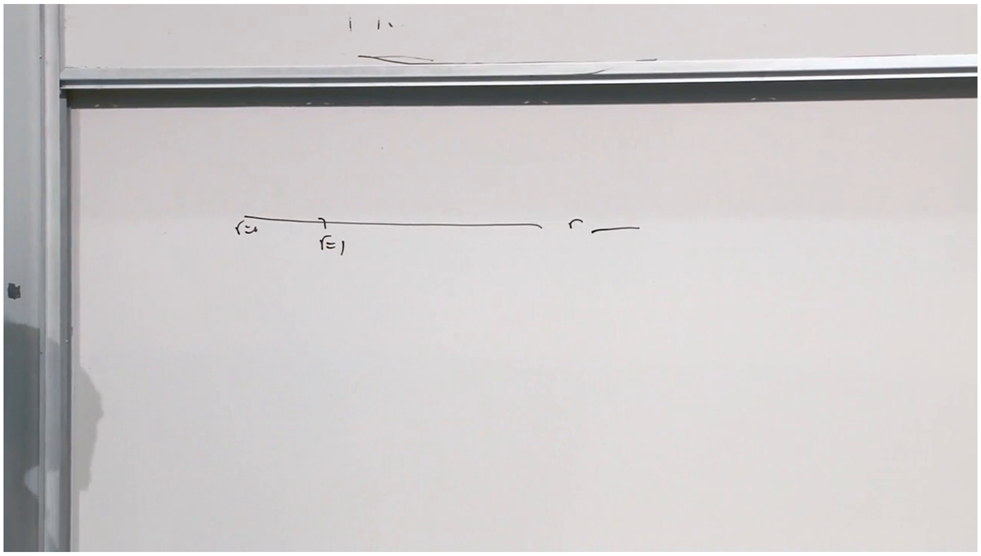

Figure 1.1: The radial coordinate line used on the board. The horizon is at $r = 1$, while $r = 0$ is the curvature singularity.

Lecture frame: 00:16:11.

Thus the complicated radial term becomes exactly $-d\rho^2$. That simplification is the reason for introducing ρ.

Now rescale the time coordinate once more,

$$\omega = \frac{t}{2}. \tag{1.9}$$

Since r is a function of ρ, the metric can be written

$$d\tau^2 = F(\rho)\rho^2 d\omega^2 - d\rho^2 - r(\rho)^2 d\Omega_2^2. \tag{1.10}$$

Question from the audience. Is $F(\rho)$ simply being defined so that the time coefficient has the form $F(\rho)\rho^2$?

Response. Yes. At this point the factorization has no independent physical content. It is a convenient definition. Its usefulness appears when the limits of F are taken near and far from the horizon. [a]

[a] Lecture timestamp: 00:26:39–00:26:59.

Far away, the Schwarzschild time coefficient tends to one. Because $dt = 2d\omega$, this becomes

$$\lim_{\rho\to\infty} F(\rho)\rho^2 = 4. \tag{1.11}$$

Near the horizon, Eq. (1.7) gives

$$\rho \simeq 2\sqrt{r-1}, \qquad r(\rho) = 1 + \frac{\rho^2}{4} + O(\rho^4), \qquad \lim_{\rho\to 0} F(\rho) = 1. \tag{1.12}$$

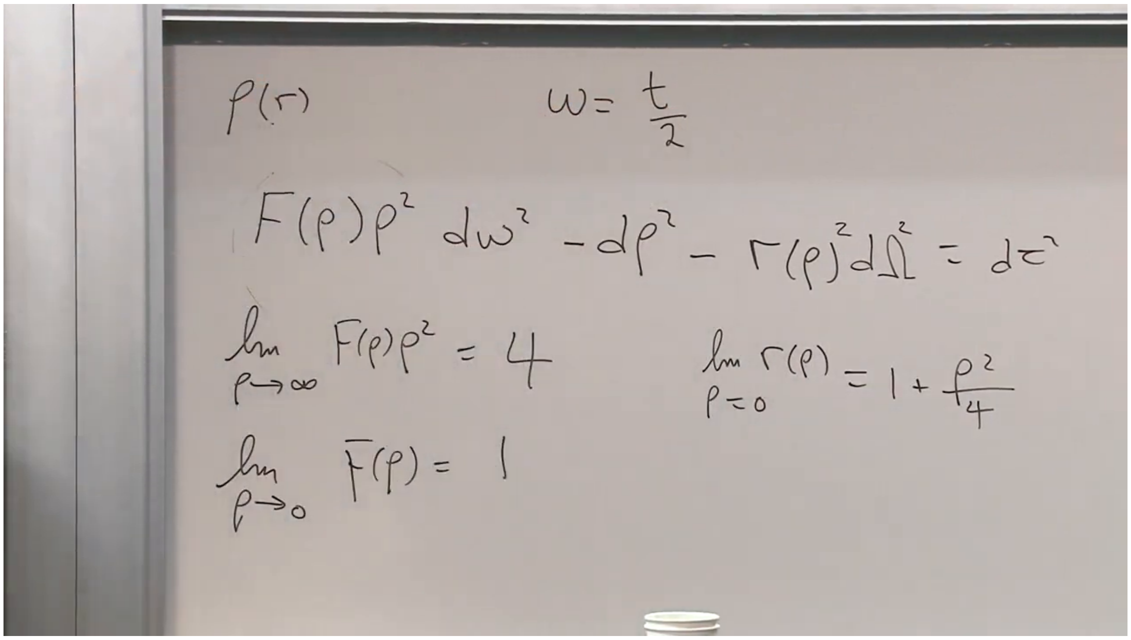

Figure 1.2: The metric in (ρ, ω), its asymptotic time coefficient, and the near-horizon limits $F \to 1$ and $r = 1 + \rho^2/4 + \cdots$.

Lecture frame: 00:37:30.

1.4 Flat Spacetime Hidden at the Horizon

Near $\rho = 0$, the radial-time part of Eq. (1.10) reduces to

$$d\tau^2_{\text{radial}} \simeq \rho^2 d\omega^2 - d\rho^2. \tag{1.13}$$

This is flat Minkowski spacetime in hyperbolic polar coordinates. Define

$$X = \rho \cosh \omega, \qquad T = \rho \sinh \omega. \tag{1.14}$$

Then

$$dT^2 - dX^2 = \rho^2 d\omega^2 - d\rho^2. \tag{1.15}$$

Constant ρ curves satisfy

$$X^2 - T^2 = \rho^2 \tag{1.16}$$

and are hyperbolae. Constant ω curves are rays from the origin. As $\omega \to \infty$, those rays approach the null line $T = X$. The Schwarzschild horizon, apparently relegated to infinite coordinate time, is therefore an ordinary null line in the regular diagram.

The small-ρ limit is unlike the small-radius limit in Euclidean polar coordinates. A small Euclidean circle contracts toward one point. A constant-ρ hyperbola approaches an entire null line. Thus $\rho = 0$ labels the whole horizon, not merely the crossing point of the axes.

The exterior formula can be analytically continued to $r < 1$. Near the horizon, $r - 1 \simeq \rho^2/4$, so the interior corresponds formally to $\rho^2 < 0$. One may call ρ imaginary, but it is cleaner to use ρ^2, r, or regular coordinates T, X. The signs of the Schwarzschild t and r coefficients exchange inside. That does not mean that a physical wristwatch becomes a ruler. It means the Schwarzschild coordinates have exchanged their causal character.

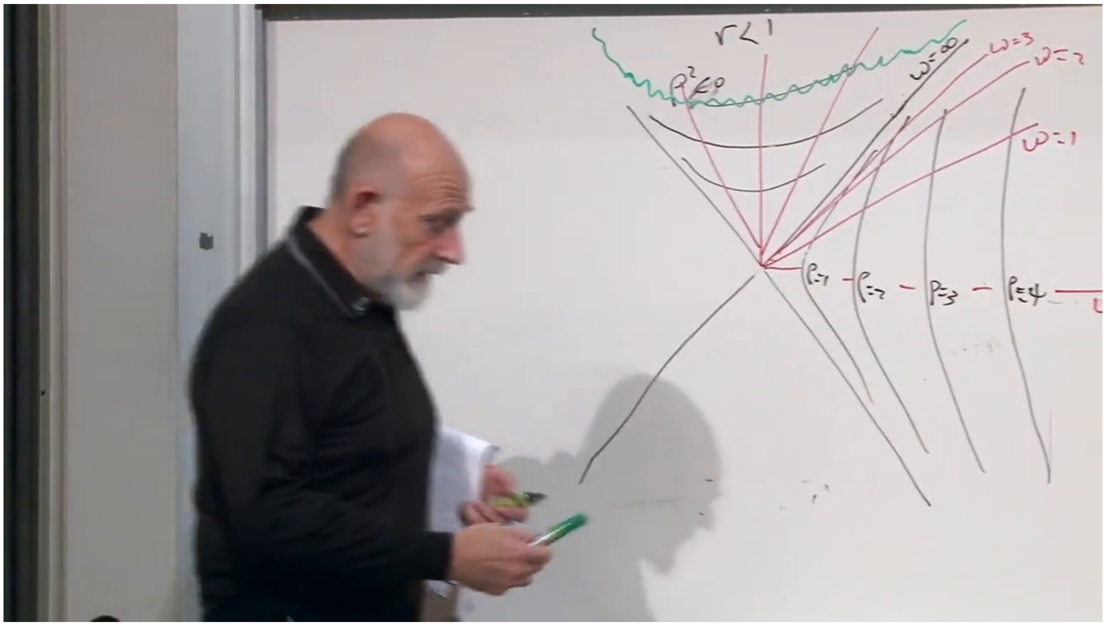

Figure 1.3: The completed board geometry: constant-ρ hyperbolae, constant-ω rays, the null horizon, and the $r < 1$ region above it.

Lecture frame: 00:38:30.

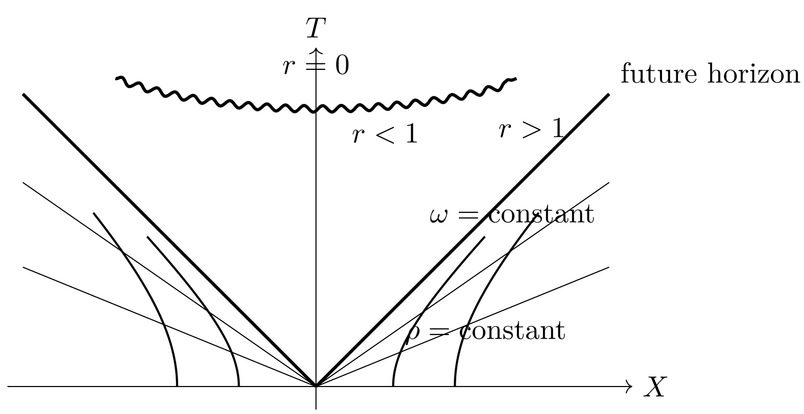

Figure 1.4: A clean causal reconstruction. The horizon is a null line, while the curvature singularity $r = 0$ is a future boundary inside the hole.

Question from the audience. What are the horizontal hyperbolae inside the horizon, and what are the straight rays?

Response. The hyperbolic curves are constant r, equivalently constant ρ^2, continued into $r < 1$. The rays are constant Schwarzschild time. Outside they form a useful coordinate grid; inside the same labels must not be mistaken for physical observer trajectories. [a]

[a]Lecture timestamp: 00:44:12–00:44:59.

Question from the audience. Is the vertical direction a time coordinate, and whose reference frame does it represent?

Response. It is a regular time direction in the new spacetime diagram, but it is not the singular Schwarzschild time t. The diagram is adapted to freely falling causal structure. An observer held at fixed exterior radius follows a hyperbola and must accelerate; a freely falling observer crosses the null horizon smoothly. [a]

[a]Lecture timestamp: 00:45:54–00:46:48.

Question from the audience. If $\rho^2 < 0$ inside, where is the explicit factor of i?

Response. One can introduce an imaginary ρ, but nothing is gained by doing so. The invariant content is carried by r, by ρ^2, and by the real regular coordinates T, X. The relation $r - 1 \simeq \rho^2/4$ is only the near-horizon expansion. [a]

[a]Lecture timestamp: 00:52:35–00:53:00.

The regular diagram is often described in the lecture as the Kruskal picture. Strictly, the exact Kruskal coordinates include a radial factor that keeps the full Schwarzschild metric regular, whereas Eq. (1.14) is its near-horizon form. The causal message is the same: the horizon is smooth, but every future-directed timelike or null curve inside ends at $r = 0$. No superluminal turn can carry it back through the horizon.

1.4.1 How the Coordinate Singularity Was Understood

The horizon's smoothness was historically difficult to see because the original Schwarzschild coordinates make its coefficients look singular. Eddington and later Finkelstein introduced coordinates that cross the horizon, and Finkelstein recognized the one-way causal boundary. Kruskal, whose facility with coordinate changes came from plasma physics, supplied a global regular picture of the eternal solution.

The lecture then pauses over John Wheeler, who helped make the phrase "black hole" unavoidable. The pause is personal as well as historical: Wheeler is remembered as a gentle and thoughtful friend even where political judgments, including judgments about nuclear competition, differed sharply. A humorous personal anecdote is offered to resist a caricature of Wheeler as socially conventional. The technical point returns in the retort that "black holes have no hair." A disturbed nonrotating hole rapidly settles toward spherical symmetry; a rotating one settles to the stationary shape fixed by its conserved parameters. This is the lecture's qualitative version of the no-hair idea, not a proof of the full theorem.

1.5 Alice Crosses; Bob Stays Outside

The maximally extended diagram contains regions that do not occur in the simple spacetime of a star collapsing from ordinary initial data. For the observer experiment we use only the exterior, the future horizon, and the black-hole interior.

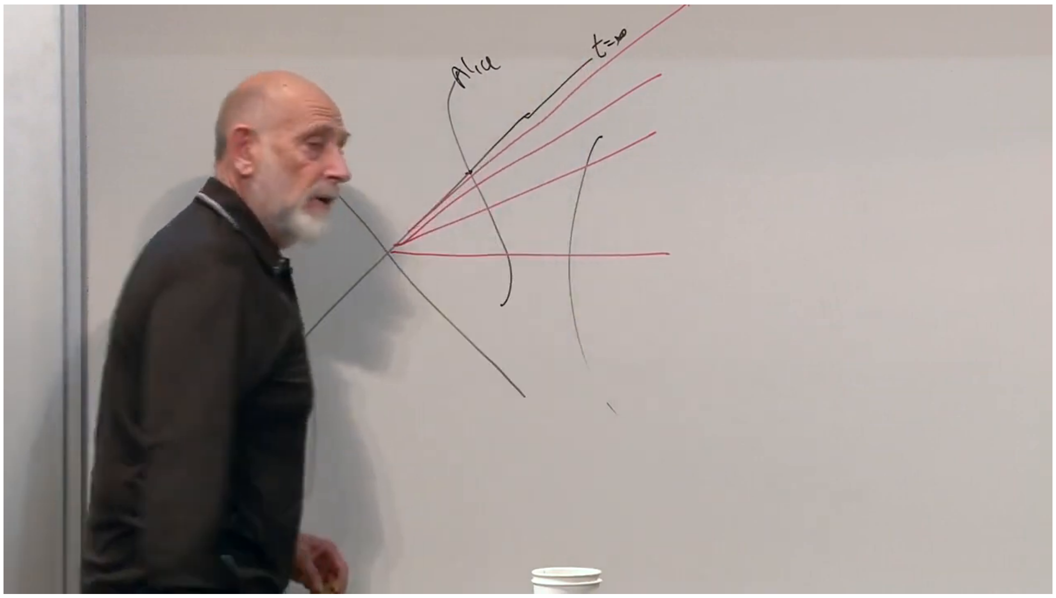

Figure 1.5: Alice's infalling worldline and the constant-Schwarzschild-time rays. The limiting ray is labelled $t = \infty$.

Lecture frame: 00:54:54.

Alice falls freely. Her worldline crosses the horizon and reaches the singularity in finite proper time. Nothing singular happens to her clock at the horizon if the black hole is large enough that tidal forces there are small.

Bob instead remains at fixed exterior radius. Holding station requires acceleration. Light from later points of Alice's trajectory reaches him after increasing delays and with increasing redshift. In Schwarzschild time, Alice approaches $t = \infty$ and Bob never receives a signal emitted after her horizon crossing. These are statements about causal signals and the outside slicing, not evidence for a material surface at the horizon.

Question from the audience. If Bob sees Alice slow down, must Alice see Bob speed up?

Response. No. Alice continues to receive ordinary light signals from Bob while her worldline lasts. There is a last event on Bob's worldline that can reach her, but that event is last only because Alice subsequently hits the singularity. Nothing abrupt happens to Bob. Observation always looks into a past light cone, and the two worldlines have an asymmetric causal relation. [a]

[a] Lecture timestamp: 00:58:31–01:01:23.

Question from the audience. If Bob knows that Alice is still close to the horizon in his coordinates, can he go down, retrieve her, and return?

Response. A rescue worldline must meet Alice before she crosses the event horizon. After that event no future-directed path can meet her and return outside. The fact that Schwarzschild slices place the crossing at $t = \infty$ does not leave Alice sitting on a retrievable material surface; it shows that those slices do not provide a global notion of "now" across the horizon. [a]

[a]Lecture timestamp: 01:01:24–01:02:12.

1.6 Charlie Follows Alice

Let Charlie fall a short distance behind Alice. Both are then approximately inertial near the horizon. They can exchange signals, converse in both directions, or even hold hands. Neither finds a signpost marking the horizon. The contrast with Bob is not a contrast between three named people. It is a contrast between free fall and an accelerated worldline that remains outside.

Question from the audience. What does Bob see if Alice falls first and Charlie follows, like a basketball following a hoop?

Response. Bob sees both approach the horizon. Alice, being ahead, appears progressively slowed and radially compressed first. Charlie appears closer and closer behind her but never catches her in Bob's limiting outside view. The visual piling-up does not describe what either infaller feels. [a]

[a]Lecture timestamp: 01:04:12–01:06:05.

Question from the audience. Does the apparent flattening contradict the tidal force, which should stretch an infaller radially?

Response. They are different effects. The flattening belongs to Bob's Schwarzschild-coordinate description. Tidal elongation is a local curvature effect that Alice and Charlie can feel. The observer discussion assumes a sufficiently large black hole, or sufficiently small observers, so that the tidal effect at the horizon is negligible. A roughly solar-mass hole would instead have fierce horizon tides, while a supermassive one can have weak horizon tides. [a]

[a]Lecture timestamp: 01:06:07–01:08:18.

Question from the audience. Does Charlie see Alice cross while he is still outside?

Response. Draw the null ray. The light emitted at Alice's crossing runs along the horizon and reaches Charlie when Charlie crosses. At that event he can calculate that he has seen her crossing, although neither observer encounters a local marker. Afterward he can see her inside. If he accelerates away before crossing and succeeds in remaining outside, he never sees the crossing. [a]

[a]Lecture timestamp: 01:08:19–01:11:11.

Question from the audience. If Alice is already inside while Charlie is outside, how can their later radii and crossing times be compared?

Response. There is no observer-independent simultaneity relation to read off from a phrase such as "at the same time." Alice sees Charlie on her past light cone and can see his later crossing only when she is farther inside. The specific worldlines and null rays answer the question. A broad extrapolation to many widely separated infallers must be checked by drawing their actual causal diagram rather than assuming one universal crossing time. [a]

[a]Lecture timestamp: 01:11:14–01:12:31.

In the regular near-horizon coordinates, a free-fall trajectory is nearly a straight line. It is not exactly straight in the full Schwarzschild geometry, but the approximation becomes good near a large, weakly curved horizon.

Question from the audience. Can string theory settle what happens in the quantum-gravity regime of the black-hole interior?

Response. That is precisely where the subject becomes subtle. A quantum theory of gravity is required near the singularity, and string theory is a candidate framework, but the relevant questions were active and difficult rather than solved by a simple application of the classical diagram. [a]

[a]Lecture timestamp: 01:13:04–01:13:38.

Question from the audience. What happens to Alice's own clock and velocity as she crosses?

Response. Her clock keeps ticking at its ordinary local rate and records finite proper time. In her instantaneous rest frame she is at rest; relative to a local freely falling frame she has an ordinary finite velocity. Near a large horizon the geometry is almost flat, so equal proper-time marks along her worldline remain regular until curvature becomes strong near the singularity. [a]

[a]Lecture timestamp: 01:13:39–01:14:44.

Question from the audience. What if Bob accelerates outward but not quite hard enough to avoid falling through?

Response. Then his worldline bends outward but still crosses the horizon. If he crosses very late, he can see Alice shortly after her crossing, but his remaining proper time before the singularity is short. Worldline slopes on the sketch are not calibrated rulers or clocks; the causal ordering, not their drawn Euclidean lengths, is what survives. [a]

[a]Lecture timestamp: 01:14:47–01:16:31.

All of these conclusions assume that no signal outruns light. The lecture briefly jokes about then-recent press reports of apparently superluminal neutrinos in Italy, an allusion to an experimental timing error rather than an exception to relativity. The causal construction itself remains bounded by the null lines.

Question from the audience. Can synchronized clocks carried by Alice and Bob decide the issue if Alice dives toward the horizon and returns?

Response. They may compare clocks when they coincide, but after their worldlines separate there is no unique distant synchronization, especially when one changes motion. Draw both worldlines and the exchanged null rays. That diagram supplies the operational answer to any specified experiment. [a]

[a] Lecture timestamp: 01:17:34–01:18:25.

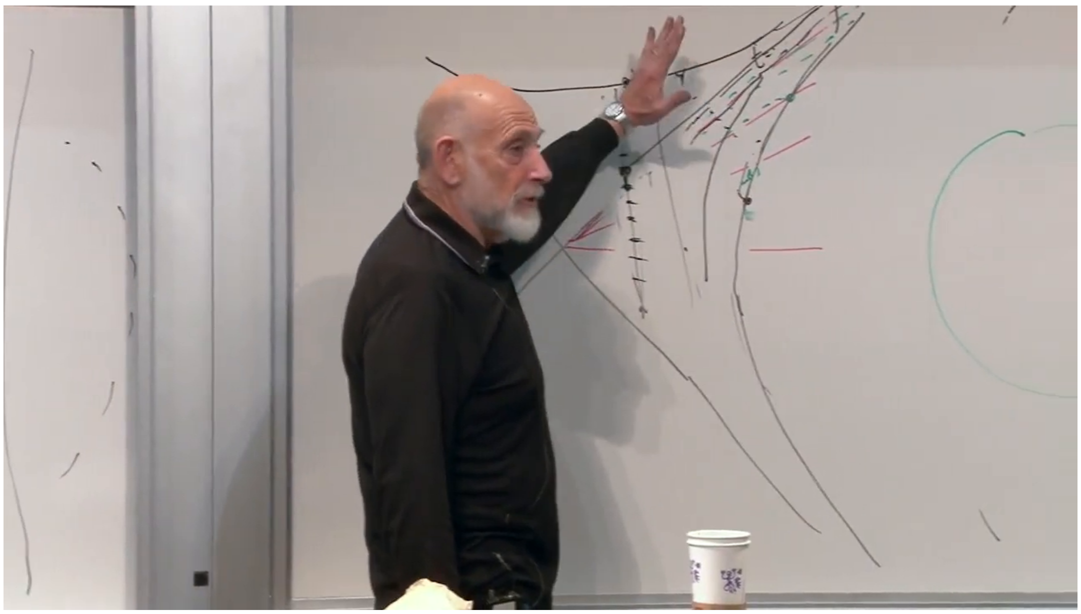

Figure 1.6: The accumulated causal diagram used to answer the Alice, Bob, and Charlie questions. Each observation is located by tracing a null ray from the observed event to the observer's worldline.

Lecture frame: 01:18:12.

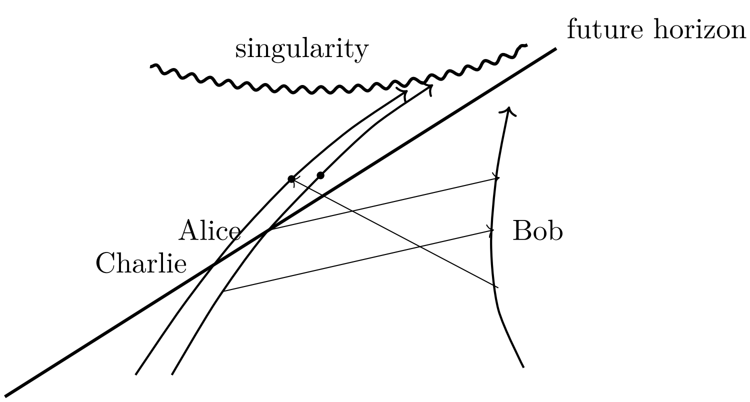

Figure 1.7: A schematic observer diagram. Bob's exterior observations and the mutual observations of Alice and Charlie are different null-ray constructions, not reciprocal clock slogans.

Question from the audience. Does Bob see Alice for an infinite time, and does that mean she never dies in his reality?

Response. Bob receives her signals over an unbounded interval of his time, but he sees her clock slow toward zero rate and receives only the finite sequence of heartbeats emitted before her crossing. The word "reality" hides the distinction between a local event and an observer's asymptotic record. Alice experiences no stopped clock at the horizon; Bob's record is increasingly delayed and redshifted. [a]

[a]Lecture timestamp: 01:18:32–01:20:59.

Question from the audience. Does Alice suddenly see all the old matter that appeared frozen on the horizon once she crosses?

Response. No sudden clearing occurs. As she falls, the past light cones by which she sees earlier matter change continuously. She can infer that light reaching her now was emitted earlier and that the matter crossed before her. Charlie's view of Alice supplies the same lesson: neither crossing produces a local discontinuity or reveals a hidden pile of chairs and dinosaurs. [a]

[a]Lecture timestamp: 01:21:02–01:23:48.

Question from the audience. Are there realistic simulations of what an infaller would see?

Response. The lecture points to Andrew Hamilton's black-hole visualizations. They can convey the severe aberration and frequency shifts of incoming light, although the display may be more disorienting than illuminating. "Nothing special at the horizon" refers to local geometry, not to an ordinary-looking sky. [a]

[a]Lecture timestamp: 01:24:11–01:25:29.

1.7 From an Eternal Hole to a Growing Hole

The lower apex of the maximally extended eternal diagram has little direct significance for a black hole formed by stellar collapse. Only the portion containing the exterior, the future horizon, and the future interior appears in the simple collapse geometry. This distinction becomes essential when matter is allowed to change the mass of the hole.

Question from the audience. What physical event is represented by the lower apex of the idealized diagram?

Response. For the eternal mathematical solution it is part of the maximal extension. A black hole formed from collapsing matter does not contain that whole lower region. Its horizon has a formation history, which requires a different global diagram and is developed in the next lecture. [a]

[a]Lecture timestamp: 01:25:59–01:26:52.

When a small test body falls in, its backreaction can be neglected and the fixed Schwarzschild diagram is adequate. If a substantial mass falls in, the black-hole mass and Schwarzschild radius increase:

$$r_S = 2GM, \qquad \Delta r_S = 2G\,\Delta M. \tag{1.17}$$

The horizon deforms toward the incoming object, the horizons join, and the distorted common horizon rapidly settles. In the lecture's nonrotating idealization the endpoint is a slightly larger sphere. A real merger also radiates energy in gravitational waves and generally leaves a rotating hole.

Question from the audience. If the horizon grows when mass falls in, does it move outward to meet matter that seemed frozen near the old horizon?

Response. For a sufficiently massive infalling object the test-particle picture is no longer valid. The horizon deforms, joins the incoming object's trapped region, and relaxes to a larger stationary horizon. The exact placement of the event horizon is global and can extend into a region before the infalling matter arrives. The lecture deliberately defers the full construction to the next meeting. [a]

[a]Lecture timestamp: 01:26:55–01:29:05.

Question from the audience. Why can astronomers see flashes when matter is said to take infinite Schwarzschild time to reach the horizon?

Response. The bright radiation does not come from the horizon itself. Gas in an accretion flow or disk collides and heats outside the horizon, including near strong photon orbits. That exterior plasma can emit light that reaches us. The empty Schwarzschild geometry isolates the causal problem but is not a complete model of an astrophysical environment. [a]

[a]Lecture timestamp: 01:29:07–01:30:22.

The characteristic relaxation time is set by the light-crossing time:

$$t_{\rm cross} \sim \frac{r_{\rm S}}{c}, \qquad t_{\rm relax} \sim \text{a few to a few tens of } t_{\rm cross}. \tag{1.18}$$

The lecture estimates 10^{-5} s as the crossing time of a kilometer-scale hole and roughly 10^{-3} s for a small merger to become nearly stationary. For a billion-solar-mass hole the corresponding time is measured in minutes to hours. These are scaling estimates, not precision merger calculations. The geometric sequence is the point: dumbbell, peanut, nearly spherical, stationary.

Question from the audience. How fast does a distorted or newly merged horizon settle?

Response. On a time scale not vastly longer than light needs to cross the hole. Because every length and time scales with $r_{\rm S}$, small holes settle in fractions of a second and supermassive holes on much longer clock times. The no-hair statement describes the stationary endpoint; the transient deformation carries away energy through gravitational radiation. [a]

[a]Lecture timestamp: 01:30:26–01:33:18.

Question from the audience. What does Bob see if earlier matter lies between the old and new horizon while another heavy body falls in?

Response. That question cannot be answered consistently by patching two static Schwarzschild diagrams together. One must solve or construct the time-dependent collapse geometry and locate its global horizon. The lecture keeps the question and explicitly postpones the answer to the next week, where horizon formation is the subject. [a]

[a]Lecture timestamp: 01:33:20–01:35:41.

1.8 What Changes When the Scale Changes?

The dimensionless diagram does not change when r_{S} changes. The physical meaning of a drawn interval does:

$$d\tau_{\text{physical}}^2 = r_{\mathrm{S}}^2\, d\hat{\tau}^2, \qquad \ell_{\text{physical}} = r_{\mathrm{S}}\, \hat{\ell}. \tag{1.19}$$

A dimensionless separation that represents half a kilometer for a kilometer-scale hole represents roughly half a billion kilometers for a hole a billion times larger. The analogy with equally drawn spheres returns: the shape is unchanged, while the number of meter sticks represented by a line scales with the radius.

Question from the audience. How does the causal diagram change as the Schwarzschild radius changes?

Response. It does not change after the radius has been scaled out. Every physical length and time assigned to the diagram is multiplied by r_{S}, while curvature at corresponding dimensionless points is multiplied by r_{S}^{-2}. [a]

[a] Lecture timestamp: 01:35:41–01:38:10.

Question from the audience. Does the causal conclusion change if Bob orbits instead of hovering at one angular position?

Response. The detailed redshifts and arrival directions become more complicated, but the horizon's causal separation does not change. No exterior timelike orbit receives a signal emitted from inside the horizon. [a]

[a] Lecture timestamp: 01:38:12–01:38:26.

Question from the audience. Since gravity vanishes inside a spherical shell, does it vanish inside the horizon because the black-hole mass is spread over that surface?

Response. No. The event horizon is not a material shell carrying the mass. The Newtonian shell theorem therefore does not apply to it. The lecture compares the static exterior solution to a point source at the curvature singularity, but matter currently falling during formation requires the time-dependent geometry deferred to the next lecture. [a]

[a] Lecture timestamp: 01:38:28–01:40:41.

1.9 Rotation and Radial Geometry

Angular momentum survives collapse. A rotating black hole has an exterior geometry different from Schwarzschild: its stationary solution is Kerr, rotation distorts the horizon geometry, and frame dragging carries inertial directions around the hole. In principle the angular momentum is measured from those exterior gravitational effects. Matter with angular momentum does not lose it merely by crossing a horizon.

Question from the audience. Can a black hole have measurable angular momentum, and does rotation change gravity outside it?

Response. Yes. The angular momentum remains as a conserved parameter, deforms the stationary geometry, and produces frame dragging. The horizon is no longer the simple spherical Schwarzschild horizon. The lecture compares the distortion qualitatively with the Earth's oblate shape, while noting that the full rotating solution is substantially more complicated. [a]

[a] Lecture timestamp: 01:40:45–01:42:36.

Question from the audience. Does the relation between r and proper distance ρ mean that gravity contracts space?

Response. The precise statement is $d\rho/dr = \sqrt{r/(r-1)} > 1$ outside the unit horizon: a given coordinate increment dr corresponds to a larger proper radial interval near the horizon. Saying that space is contracted or stretched depends on which coordinate intervals are being compared. The invariant statement is the proper-distance relation itself. [a]

[a] Lecture timestamp: 01:42:36–01:43:20.

1.10 Final Questions

Question from the audience. From Bob's view Alice's clock approaches a stop. From Alice's view, does Bob's space or time come to an end?

Response. No. Alice sees ordinary signals from successive events on Bob's worldline until her own worldline ends. She fails to see later Bob events because no later point on her worldline exists, not because Bob or his space disappears. In these regular causal coordinates radial null rays are drawn straight; their appearance as curved paths in another coordinate system does not alter which events they connect. [a]

[a] Lecture timestamp: 01:43:21–01:45:20.

General relativity presents two complementary tasks. Given a metric, one can trace geodesics and null rays to determine what observers measure. Given a distribution of matter, one must solve Einstein's equations to find the metric. The first task has occupied this lecture. The second is usually harder, although symmetric collapse provides tractable examples.

Question from the audience. What happens to the horizon and singularity of an evaporating black hole?

Response. The lecture gives a schematic picture in which the horizon shrinks and meets the endpoint of the singular region. Evaporation is a semiclassical quantum process, and the final Planck-scale endpoint is not settled by the classical Schwarzschild diagram. The shrinking sketch should therefore be read as a qualitative preview, not a complete quantum geometry. [a]

[a] Lecture timestamp: 01:46:04–01:46:32.

Almost every astrophysical black hole should rotate. Stars already possess angular momentum, and collapse amplifies their angular velocity in the familiar manner of a skater drawing in their arms. The singular structure of a Kerr black hole is not the Schwarzschild $r = 0$ surface drawn here, and it should not be described as matter literally rotating with infinite linear speed.

Question from the audience. Are nonrotating macroscopic black holes common, and how many black holes are there?

Response. Exactly zero angular momentum would be an exceptional idealization. The lecture gives only an order-of-magnitude classroom guess for the cosmic population, somewhere between galaxy and stellar counts and perhaps around 10^{15}. That is a lecture-era estimate, not a measured census. The robust point is that collapse generically preserves and amplifies some angular momentum. [a]

[a]Lecture timestamp: 01:46:32–01:48:42.

Question from the audience. Do black holes form from existing matter or appear through a quantum process?

Response. Ordinary astrophysical holes form when existing stellar matter collapses. Nuclear burning and the outward flow of energy help support a star. When fuel is exhausted, the remnant contracts. Depending on its mass and available pressure support, it may become a white dwarf, a neutron star, or, if no support can halt collapse, pass within its own Schwarzschild radius and form a black hole. [a]

[a]Lecture timestamp: 01:48:43–01:50:23.

Question from the audience. Could microscopic black holes form, and what would set their lifetime and minimum size?

Response. This part is speculative. Hawking evaporation is faster for smaller mass, with a lifetime that scales approximately as M^3. The lecture uses a mountain-mass comparison for a hole whose lifetime might be cosmological; the numerical analogy is only rough. Violent density fluctuations in the early universe or sufficiently energetic collisions are conceivable formation mechanisms. A black hole near the Planck mass, roughly $10^{-5}\,\mathrm{g}$, marks the scale where a classical geometric description ceases to be trustworthy. Reaching that energy in an ordinary accelerator would require an implausibly large machine under straightforward extrapolation, although the formation statement is possible in principle. [a]

[a]Lecture timestamp: 01:50:23–01:53:31.

Question from the audience. Will the course reach cosmology, dark matter, dark energy, entropy, and gravitational waves?

Response. Cosmological applications and perhaps gravitational waves remain possible later topics. Black-hole entropy is deferred beyond the quarter. The lecture closes by accepting that only about a quarter of the planned material was covered that evening: resolving the questions carefully matters more than racing through the outline. [a]

[a]Lecture timestamp: 01:53:33–01:54:38.

The operational rule survives every question in the room. Specify the worldlines, draw the null rays, and locate the events they connect. The horizon is not a place where local physics breaks; it is the boundary beyond which every future-directed signal remains inside. The next lecture replaces the eternal background by a collapsing source and shows how that boundary is formed.